ARCUS

INFRASTRUCTURE INTELLIGENCE

PATH 01 - NEW TERRITORY

Territory Briefing: Torino

Professional province-level preliminary briefing: historical evidence, Atlas map, project-context focus and operational next checks.

INFRASTRUCTURE INTELLIGENCE / GENERATED 08/06/2026

REPORT ID	VERSION	ANALYSIS	SPATIAL LEVEL	GENERATED BY
ARCUS-P01-20260608-torino	ARCUS Professional PDF-ready v0.3	Territory Briefing	Province level	ARCUS Professional

01 EXECUTIVE SUMMARY

ARCUS identifies 42 documented cases in the selected province and translates the Step 3 exposure indicators into operational actions for Bridge interventions. The PDF map shows ARCUS cases; the hazard-layer contribution is summarised through the indicators below. The dominant territorial signal is **Hydraulic**; the leading historical driver is **Hydraulic**.

Recommended use: province-level preliminary screening for Bridge interventions, before design, due diligence or site-specific investigation.

FINDING 1

The hydraulic context dominates the territorial reading.

FINDING 2

41 triggered events and 30 total collapses shape the technical priority.

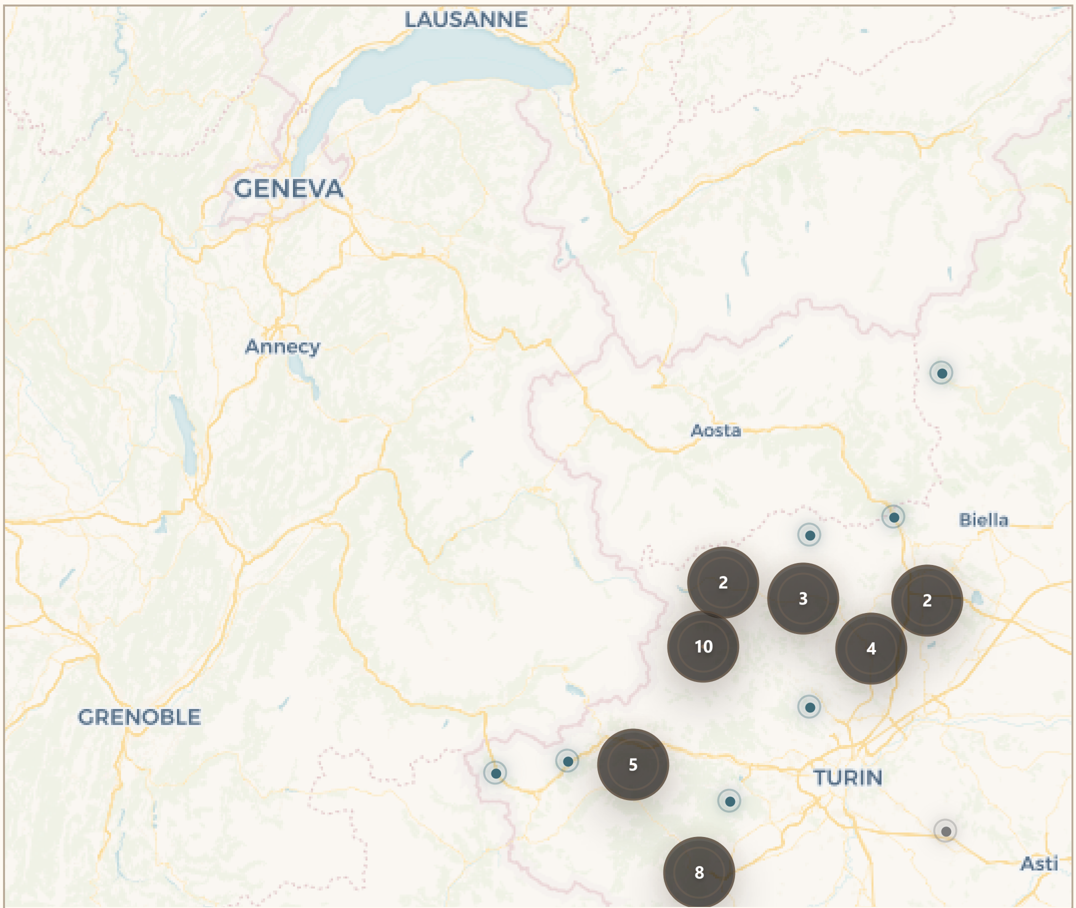
FINDING 3

Very high evidence reliability with 125 linked sources.

WHY ARCUS FLAGS THIS

- Dominant exposure: Hydraulic (97 / 100).
- Historical severity: 30 total collapses and 41 triggered events.
- Map-linked cases: P1 Castellamonte, P2 Ala di Stura, P3 Ala di Stura.
- Evidence package: 125 linked sources supporting the signal.

02 SELECTED PROVINCE & ARCUS CASES MAP



The PDF map shows ARCUS georeferenced cases within the selected province. Hazard overlays are not visually stacked to preserve readability; their contribution is summarised through the exposure indicators below. Full interactive layer control is available in ARCUS Professional Atlas.

> PDF map = ARCUS cases map. Exposure indicators = hazard overlay synthesis. Professional Atlas = full interactive layer exploration.

03 HOW TO READ THIS REPORT

ARCUS starts from the selected province, reads all records available for that administrative area, identifies the dominant territorial pattern and turns that signal into operational next checks. This is province-level screening, not structural safety certification.

SELECTED PROJECT CONTEXT

Bridge

ARCUS reading: for Bridge interventions, the Hydraulic signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Torino.

DRIVER

Hydraulic / 97

The selected province is mainly driven by hydraulic exposure: 41 matched collapses out of 42 analysed events (98%). Other layers should be read as secondary preliminary-screening context.

04 KEY INDICATORS

PRIORITY INDEX

82 / 100

HIGH ATTENTION

Hydraulic

EVIDENCE RELIABILITY

96 / 100

VERY HIGH

125 linked sources

HISTORICAL FAILURE CONTEXT

72 / 100

HIGH

Hydraulic

HISTORICAL EVENTS

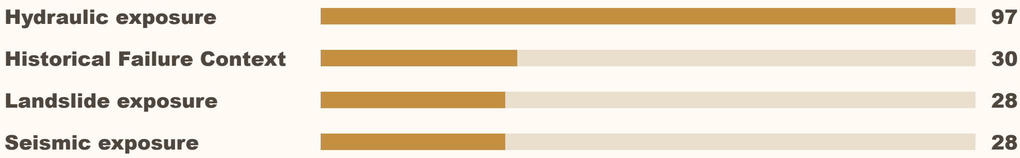
42

98% HYDRAULIC

41 triggered events

05 HAZARD / EXPOSURE CONTEXT

The Step 3 analysis combines public Hydraulic, Landslide and Seismic layers with ARCUS historical cases. For PDF readability, the map displays ARCUS cases only. The influence of the active hazard layers is represented through the exposure scores below.



06 OPERATIONAL MEANING

PRIMARY FOCUS

Preliminary hydraulic compatibility; Scour susceptibility; Foundation, pier and abutment exposure

ARCUS reading: for Bridge interventions, the Hydraulic signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Torino.

ATTENTION POINTS

- Preliminary hydraulic compatibility
- Scour susceptibility
- Foundation, pier and abutment exposure
- Riverbed dynamics and sediment/debris transport
- Flood/debris scenarios and inspection accessibility

07 PRIORITY ARCUS CASES

The selected province contains all ARCUS records available for that administrative area. This section highlights the highest-priority cases within the provincial set, ranked by severity, trigger, dominant driver and evidence reliability. These cases are not external comparables: they explain the dominant territorial signal used by the report.

P1	**B00.10.15 - Castellamonte**	**2000-10-13 / Hydraulic**	**Total Collapse / A**
P2	**B00.10.03 - Ala di Stura**	**2000-10-13 / Hydraulic**	**Total Collapse / A**
P3	**B00.10.05 - Ala di Stura**	**2000-10-13 / Hydraulic**	**Total Collapse / A**

08 TERRITORIAL PATTERN INTERPRETATION

The selected province shows a concentrated Hydraulic pattern. This pattern supports preliminary attention toward preliminary hydraulic compatibility, scour susceptibility, foundation, pier and abutment exposure, riverbed dynamics and sediment/debris transport for Bridge interventions.

09 INTERPRETATION

For new Bridge interventions, ARCUS directs first checks toward Hydraulic. The reading is derived from provincial ARCUS cases, hazard overlays and evidence quality: it does not automatically indicate structural criticality, but defines what should be checked next.

10 IMMEDIATE SCREENING ACTIONS

ACTION 1

Use the Step 3 exposure indicators to identify the dominant Hydraulic signal.

ACTION 2

Review P1-P3 Priority ARCUS Cases.

ACTION 3

Request the minimum technical data package.

ACTION 4

Commission site-specific hydraulic and geotechnical checks.

11 DATA TO REQUEST BEFORE DESIGN DECISIONS

DATA 1

Hydraulic model or preliminary hydraulic study

DATA 2

Riverbed survey / bathymetry where relevant

DATA 3

Geotechnical investigation and foundation concept

DATA 4

Historical flood levels

DATA 5

Inspection accessibility constraints

12 RECOMMENDED CLIENT PATH

1. Use the Step 3 exposure indicators to identify the dominant Hydraulic signal.
2. Review P1-P3 Priority ARCUS Cases.
3. Request the minimum technical data package.
4. Commission site-specific hydraulic and geotechnical checks.
5. Use CSV and GeoJSON exports for technical coordination.

13 NATIONAL BENCHMARK

Compared with the national ARCUS baseline, the selected province is above average for Triggered-event share, Professional-grade sources, Historical Failure Context. If triggered-event share is above average, the briefing should be used to anticipate hydraulic, geotechnical and territorial-context checks before site-specific design.

14 DATA COVERAGE & LIMITATIONS

- Province-level screening: not design scale.
- ARCUS does not certify structural safety or asset condition.
- Public overlays guide priorities and must be followed by site-specific checks.

A SOURCE APPENDIX

Main sources linked to events in the selected province, deduplicated by title or URL. Total linked sources in this briefing: 125. The full source table can be exported separately from ARCUS Professional.

B00.10.15, B00.10.03, B00.10.05 +34 MORE / ARPA PIEMONTE
Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale
arpa.piemonte.it / PDF

B00.10.15, B00.10.03, B00.10.05 +34 MORE / ANSA
10 anni fa alluvione Nord-Ovest. fu inferno d'acqua – Piemonte e Valle d'Aosta
ansa.it / HTML

B00.10.15, B00.10.03, B00.10.05 +37 MORE / JOURNAL PAPER
Bridge collapses in Italy across the 21st century: survey and statistical analysis
doi.org

B05.11.01 / CORRIERE DELLA SERA
Bloccata la linea ferroviaria Torino-Roma
corriere.it / SHTML

B05.11.01 / GAZZETTA D'ASTI
Crollo della passerella a Villanova: condanne in Tribunale
gazzettadasti.it

B08.05.02 / REGIONE PIEMONTE
Evento alluvionale 29-30 maggio 2008 – relazione tecnica ufficiale
regione.piemonte.it / PDF

B08.05.02 / METEONETWORK
Alluvione maggio 2008 in Provincia di Cuneo: resoconto completo
meteonetwork.it / PDF

B17.06.01 / TORINOTODAY
Cedimento strutturale: è crollato lo storico ponte di Strambino
torinotoday.it / HTML

15 METHODOLOGY SNAPSHOT

ARCUS combines historical collapse evidence, source reliability, spatial relevance, severity, triggers, cause similarity, territorial hazard context and public WMS overlays. The briefing supports preliminary screening and prioritisation: it is not a structural safety certification.

COMPONENT	USE IN THE BRIEFING
Historical evidence	Documented ARCUS records, causes, triggers and severity.
Source reliability	Number, role and quality of available sources.
Spatial relevance	Selected province and administrative provincial boundary.
Hazard context	Public hydraulic/landslide WMS overlays and seismic context.
Priority ARCUS cases	Cases inside the selected province, ranked by severity, trigger, dominant driver and evidence reliability.

16 SCORE AND CLASS LEGEND

CLASS	INTERPRETATION
A	High reliability: multiple sources, preferably institutional or technical.
B	Medium reliability: documented event with limited coverage or source variety.
C	Limited reliability: local, incomplete or to-be-strengthened evidence.
Critical	Maximum attention: strong historical and/or territorial evidence.
High	High attention: significant evidence but not necessarily complete.
Medium / Low	Moderate or low attention, to be read together with available data.